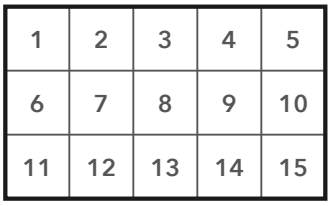
Area

Area is how many squares cover a shape.

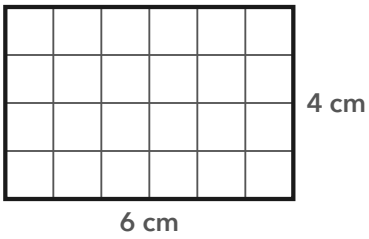
Cover it with squares, then count



Area = 15 square units

Each square is 1 square unit.

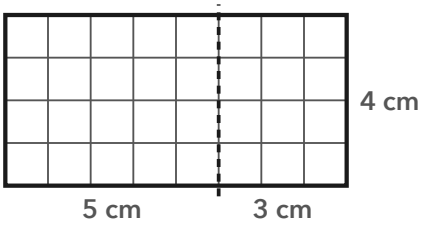
Rows × columns: no need to count one by one



4 × 6 = 24

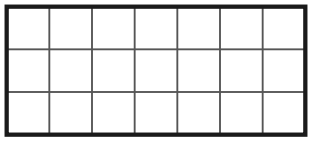
4 rows of 6 squares: 24 square cm.
Multiply the side lengths.

Break it apart: 4 × 8 is 4 × 5 and 4 × 3



4 × 5 = 20
4 × 3 = 12
20 + 12 = 32 square cm

Your turn!



3 × 7 = 21 square units

For teachers

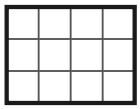
Goal: area as unit squares that cover a shape, then rows × columns (3.MD.5–7). Say "How many rows? How many in each row?" Area is inside; the edge is perimeter (packet 45). Link breaking apart to packet 34.

Area: Practice

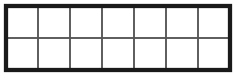
Count the squares, or multiply rows × columns.

Part 1 Count the squares Each square is 1 square unit.

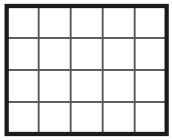
1) Area = 12 square units



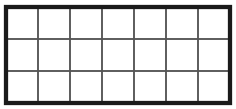
2) Area = 14 square units



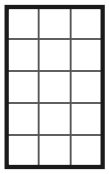
3) Area = 20 square units



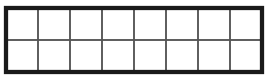
4) Area = 21 square units



5) Area = 15 square units

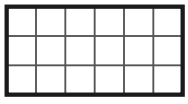


6) Area = 16 square units

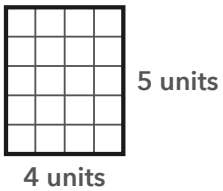


Part 2 Rows times columns Multiply instead of counting.

7) 3 × 6 = 18 square units



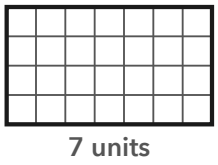
8) 5 × 4 = 20 square units



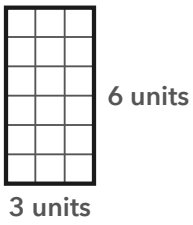
Area: Practice (continued)

Part 2 Rows times columns (continued)

9) 4 × 7 = 28 square units



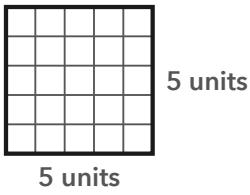
10) 6 × 3 = 18 square units



11) 2 × 9 = 18 square units

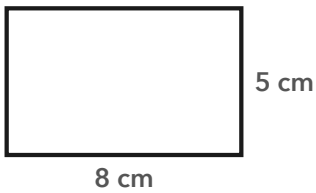


12) 5 × 5 = 25 square units

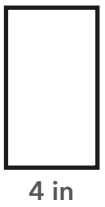


Part 3 Use the side lengths No squares drawn this time. Multiply the sides.

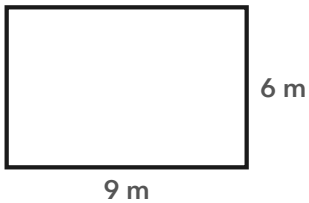
13) 5 × 8 = 40 square cm



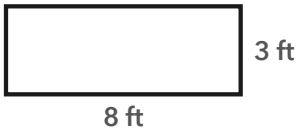
14) 7 × 4 = 28 square in



15) 6 × 9 = 54 square m



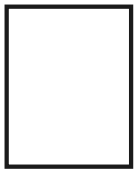
16) 3 × 8 = 24 square ft



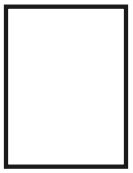
Area: Practice (continued)

Part 3 Use the side lengths (continued)

17) 9 × 7 = 63 square cm

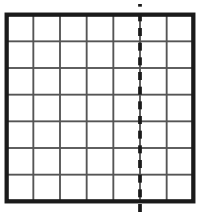


18) 8 × 6 = 48 square in



Part 4 Break it apart Find the area of each part. Add them.

19) 7 × 7 = 49

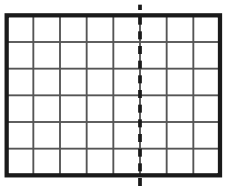


7 × 5 = 35

7 × 2 = 14

35 + 14 = 49

20) 6 × 8 = 48

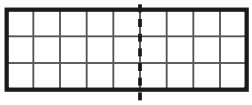


6 × 5 = 30

6 × 3 = 18

30 + 18 = 48

21) 3 × 9 = 27



3 × 5 = 15

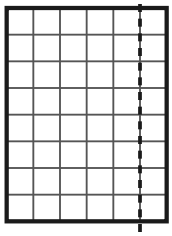
3 × 4 = 12

15 + 12 = 27

Area: Practice (continued)

Part 4 Break it apart (continued)

22) $8 \times 6 =$ 48



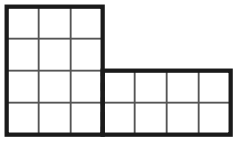
$8 \times 5 =$ 40

$8 \times 1 =$ 8

40 + 8 = 48

Part 5 Two rectangles Find each area. Add them.

23)

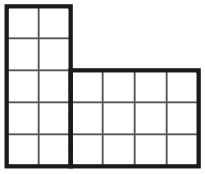


4 × 3 = 12

2 × 4 = 8

12 + 8 = 20

24)

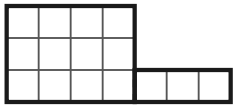


5 × 2 = 10

3 × 4 = 12

10 + 12 = 22

25)



3 × 4 = 12

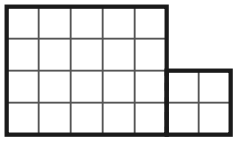
1 × 3 = 3

12 + 3 = 15

Area: Practice (continued)

Part 5 Two rectangles (continued)

26)



4 × 5 = 20

2 × 2 = 4

20 + 4 = 24

Part 6 Rectangles No help this time. Multiply the sides.

27) 6 cm by 7 cm: area = 42 sq cm

28) 8 cm by 3 cm: area = 24 sq cm

29) 9 cm by 4 cm: area = 36 sq cm

30) 5 cm by 6 cm: area = 30 sq cm

31) 7 cm by 8 cm: area = 56 sq cm

32) 3 cm by 9 cm: area = 27 sq cm

33) 5 cm by 9 cm: area = 45 sq cm

34) 2 cm by 8 cm: area = 16 sq cm

35) 4 cm by 8 cm: area = 32 sq cm

36) 9 cm by 8 cm: area = 72 sq cm

Area: Practice (continued)

Part 7 Squares No help this time. All four sides are the same.

37) 6 m by 6 m square: area = 36 sq m

38) 3 in by 3 in square: area = 9 sq in

39) 8 cm by 8 cm square: area = 64 sq cm

40) 4 ft by 4 ft square: area = 16 sq ft

41) 9 m by 9 m square: area = 81 sq m

42) 7 in by 7 in square: area = 49 sq in

Part 8 Find the missing side No help this time. Side × side = area. What is the other side?

43) 4 cm × 6 cm = 24 sq cm

44) 5 cm × 7 cm = 35 sq cm

45) 3 cm × 9 cm = 27 sq cm

46) 6 cm × 8 cm = 48 sq cm

47) 7 cm × 4 cm = 28 sq cm

48) 9 cm × 5 cm = 45 sq cm

Area: Practice (continued)

Part 8 Find the missing side (continued)

49) 2 cm × 8 cm = 16 sq cm

50) 6 cm × 7 cm = 42 sq cm

Part 9 Find the area

51) 9 in by 6 in: area = 54 sq in

52) 10 cm by 2 cm: area = 20 sq cm

53) 8 m by 2 m: area = 16 sq m

54) 2 in by 2 in: area = 4 sq in

55) 3 cm by 10 cm: area = 30 sq cm

56) 7 in by 3 in: area = 21 sq in

57) 2 cm by 7 cm: area = 14 sq cm

58) 10 m by 8 m: area = 80 sq m

59) 9 m by 9 m: area = 81 sq m

60) 5 m by 5 m: area = 25 sq m

Area: Practice (continued)

Part 9 Find the area (continued)

61) 9 ft by 9 ft: area = 81 sq ft

62) 5 m by 4 m: area = 20 sq m

63) 9 ft by 5 ft: area = 45 sq ft

64) 10 cm by 8 cm: area = 80 sq cm

65) 5 m by 10 m: area = 50 sq m

66) 3 in by 10 in: area = 30 sq in

67) 2 ft by 7 ft: area = 14 sq ft

68) 8 in by 7 in: area = 56 sq in

69) 10 in by 10 in: area = 100 sq in

70) 2 cm by 9 cm: area = 18 sq cm

71) 10 in by 6 in: area = 60 sq in

72) 4 m by 2 m: area = 8 sq m

Area: Practice (continued)

Part 9 Find the area (continued)

73) 8 cm by 10 cm: area = 80 sq cm

74) 2 cm by 3 cm: area = 6 sq cm

Part 10 Find the missing side

75) 6 m × 10 m = 60 sq m

76) 5 m × 3 m = 15 sq m

77) 10 m × 8 m = 80 sq m

78) 2 cm × 2 cm = 4 sq cm

79) 7 cm × 5 cm = 35 sq cm

80) 4 m × 10 m = 40 sq m

81) 7 m × 4 m = 28 sq m

82) 8 m × 2 m = 16 sq m

83) 10 cm × 2 cm = 20 sq cm

84) 3 m × 10 m = 30 sq m

Area: Practice (continued)

Part 10 Find the missing side (continued)

85) 8 cm × 8 cm = 64 sq cm

86) 3 cm × 4 cm = 12 sq cm

87) 4 cm × 9 cm = 36 sq cm

88) 5 cm × 3 cm = 15 sq cm

89) 6 cm × 9 cm = 54 sq cm

90) 7 m × 2 m = 14 sq m

91) 10 m × 5 m = 50 sq m

92) 5 m × 6 m = 30 sq m

Part 11 Remember? Problems from earlier packets.

93) 8 × 60 = 480

94) Round 347 to the nearest hundred: 300

95) 9 groups of 4 = 36

96) 6 sixths = 1 whole

Area: Practice (continued)

Part 11 Remember? (continued)

97) $618 - 349 =$ **269**

98) 2 fourths = **1** half

Part 12 Word problems Draw the rectangle if it helps.

99) A rug is 7 feet long and 3 feet wide. What is its area?

Number sentence $7 \times 3 = 21$

Answer **21** square feet

Draw or show your work

100) A patio has 8 rows of square tiles with 4 tiles in each row. How many tiles cover it?

Number sentence $8 \times 4 = 32$

Answer **32** tiles

Draw or show your work

101) A poster is 9 inches long and 6 inches wide. What is its area?

Number sentence $9 \times 6 = 54$

Answer **54** square inches

Draw or show your work

Area: Practice (continued)

Part 12 Word problems (continued)

102) A room is 10 feet long and 9 feet wide. What is its area?

Number sentence $10 \times 9 = 90$

Answer **90** square feet

Draw or show your work

103) Challenge: A room is 8 feet by 6 feet. A rug covers 20 square feet. How much floor is not covered?

Number sentence $8 \times 6 = 48, 48 - 20 = 28$

Answer **28** square feet

Draw or show your work

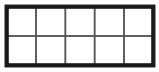
Check-In: Area

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Rows times columns

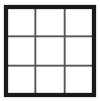
1) 2 × 5 = 10 square units



2 units

5 units

2) 3 × 3 = 9 square units

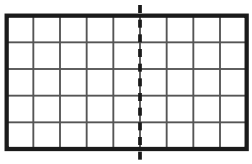


3 units

3 units

Part 2 Break it apart

3) 5 × 9 = 45



5 × 5 = 25

5 × 4 = 20

25 + 20 = 45

Part 3 Area

4) 9 ft by 4 ft: area = 36 sq ft

5) 6 in by 8 in: area = 48 sq in

6) 9 in by 8 in: area = 72 sq in

7) 6 in by 7 in: area = 42 sq in

Part 4 Find the missing side

8) 5 cm × 5 cm = 25 sq cm

9) 3 cm × 7 cm = 21 sq cm